

AIGR Deep Research Report: HSPD1/Hsp60 and GO:0045041 (Protein Import into Mitochondrial Intermembrane Space)

Gene: HSPD1 (human), UniProt P10809 **Focus type:** function_assignment **Hypothesis slug:** function-hypothesis-go-0045041 **Term under evaluation:** GO:0045041 — *protein import into mitochondrial intermembrane space*

Summary

The seed hypothesis proposes that human HSPD1/Hsp60 directly participates in protein import into the mitochondrial intermembrane space (IMS), tracing the annotation through IBA source PANTHER node PTN000143510 and yeast Hsp60 (SGD:S000004249), and asks whether the difference between the antifolding/export result

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and the stop-transfer import result
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excludes conservation in human HSPD1. After tracing the annotation provenance and reconciling the primary literature, the evidence does **not** support treating GO:0045041 as a directly demonstrated, core function of human HSPD1. It is best treated as a **weakly supported, non-core, inference-only (IBA) term**.

Three facts drive this conclusion. First, the human annotation to GO:0045041 is **IBA-only** (phylogenetically inferred from a yeast ancestor via GO_Central); there is no human experimental (IDA/IMP/EXP) support, and every experimentally supported human HSPD1 term describes **matrix chaperonin folding**, with a primary location of mitochondrial matrix. Second, the single donor experiment (yeast,

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demonstrates only an **indirect, substrate-selective matrix "antifolding" role** in conservative sorting of cytochrome b2 — not a translocase or IMS-import activity — and is **directly contradicted** on that same substrate by

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which shows the b2-presequence passenger reaches the IMS **independently of hsp60** via inner-membrane stop-transfer. Third, the **dedicated IMS import machinery is the MIA40/CHCHD4–Erv1 disulfide relay**, not Hsp60.

The most important caveat is that human HSPD1 and yeast Hsp60 are unambiguous orthologs (~58–61% identity), so an antifolding-type contribution to conservative sorting is *plausibly conserved* and the term need not be hard-refuted. But plausibility of an indirect, selective matrix chaperone contribution is a fundamentally different claim from "HSPD1 directly performs protein import into the IMS." The recommended curation posture is to **down-weight or remove GO:0045041 as a direct function, or retain it only with an explicit low-confidence / indirect-role qualifier**, while foregrounding the directly supported matrix protein-folding terms.

Key Findings

Finding 1 — The human GO:0045041 annotation rests solely on phylogenetic inference (IBA); all direct human evidence is matrix folding, not IMS import

A direct interrogation of the UniProt P10809 GO annotation set shows a clean split between what is *inferred* and what is *experimentally demonstrated* for human HSPD1. The term **GO:0045041 ("protein import into mitochondrial intermembrane space") is annotated ONLY as IBA:GO_Central** (Inferred from Biological Ancestor — a phylogenetic propagation), with **no IDA, IMP, or other experimental support**.

By contrast, every experimentally supported process/function annotation for human HSPD1 describes **matrix chaperonin activity**: protein refolding (GO:0042026, IDA), protein stabilization (GO:0050821, IMP), response to unfolded protein (GO:0006986, IDA), ATP-dependent protein folding chaperone (GO:0140662), and unfolded protein binding (implied by chaperonin function). Furthermore, UniProt's primary subcellular location for HSPD1 is

mitochondrion matrix; no IMS localization is annotated (the only membrane-associated annotation is an ISS "mitochondrial inner membrane"). This matters because a *directly demonstrated* IMS-import function would normally be accompanied by experimental evidence of the protein acting in the relevant pathway; here there is none for human. The annotation is entirely a downstream product of tree-based inference, not human experimentation.

Finding 2 — The yeast donor evidence is indirect (antifolding) and is directly contested on the same substrate (cytochrome b2)

The experimental basis for the whole inference chain is the yeast work in [PMID:1347713](#) (Koll et al., 1992, *Cell*). In yeast, cytochrome b2 reaches the IMS by a two-step "conservative sorting" route: it is first imported into the matrix, then re-exported across the inner membrane. Koll et al. showed that Hsp60 **arrests folding of the protein in the matrix ("antifolding")**, keeping the export-competent domain translocation-competent and thereby coupling import to export. This is an **indirect matrix chaperone role** — Hsp60 keeps a substrate unfolded so it can be handed to the export machinery — and is emphatically *not* an IMS translocase activity.

Critically, the same substrate is the subject of a directly contradictory result. [PMID:7911803](#) (Rospert, Müller, Schatz, Glick, 1994) analyzed a fusion protein targeted to the IMS by the **same bipartite cytochrome b2 presequence** and found it was sorted to the IMS **"independently of hsp60."** The mechanistic explanation is decisive: the b2 presequence arrests import at the inner membrane (a **stop-transfer** mechanism), so the passenger "is never exposed to hsp60." In other words, on the very substrate that motivated the Hsp60-sorting idea, the IMS route does **not** require Hsp60.

Two further papers show that even Hsp60's matrix folding role is **substrate-selective**, not general. [PMID:8631298](#) (Rospert, 1996) found that of four monomeric imported proteins, only rhodanese required hsp60 for folding; DHFR, barnase, and Cpr3p folded efficiently without it and showed no detectable binding — "the mitochondrial chaperonin system is not essential for the folding of all matrix proteins." [PMID:9774331](#) (Dubauqué et al., 1998) found in vivo hsp60/hsp10 substrates to be "overlapping but non-identical," i.e., differential and selective. Taken together, the donor evidence is (a) indirect, (b) selective, and (c) self-contradicted on its flagship substrate — a weak foundation for a directly-worded import annotation.

Finding 3 — The dedicated IMS protein-import machinery is the MIA40/CHCHD4-Erv1 disulfide relay, not Hsp60

The IMS has its **own** dedicated import pathway, mechanistically distinct from anything Hsp60 does. [PMID:33921425](#) (Dickson-Murray et al., 2021) states plainly: "The main protein import pathway for the intermembrane space (IMS) recognises proteins that are cysteine-rich, and it is the only import pathway that chemically modifies the imported precursors... The key component of this pathway is **Mia40 (called CHCHD4 in human cells)**." Precursors are trapped in the IMS by disulfide-bond formation (oxidative folding).

Supporting structural and reconstitution work reinforces this. [PMID:20136511](#) (Endo et al., 2010) describes a "dedicated disulfide relay system" consisting of Tim40/Mia40 and Erv1 that traps small soluble proteins. [PMID:19477928](#) (Tienison et al., 2009) demonstrated by in vitro reconstitution that "Mia40, Erv1, and oxygen are the minimal machinery" to import and oxidize small Tim proteins. Meanwhile, bipartite-presequence IMS proteins such as cytochrome b2 reach the IMS via the **TIM23 translocase and a stop-transfer step** (

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Neither of the two genuine IMS-import routes has Hsp60 as a core component; Hsp60's documented role is downstream, in the matrix, folding proteins *after* import.

Finding 4 — Human HSPD1 and yeast Hsp60 are clear orthologs (~58–61% identity); orthology does not exclude conservation of the antifolding role

To test whether the IBA inference could be justified on conservation grounds, a global Needleman–Wunsch alignment was performed between UniProt [P10809](#) (human HSPD1, 573 aa) and [P19882](#) (yeast Hsp60, SGD:S000004249, 572 aa). The alignment yielded **348 identical positions over 602 aligned columns (57.8% identity; 60.8% relative to the shorter sequence)**. Group I chaperonins are among the most deeply conserved protein families known, structurally and functionally.

The interpretation cuts both ways and must be stated carefully. Strong orthology means an *antifolding contribution to conservative sorting* is **plausibly conserved** in human — so the term should not be hard-refuted. But orthology to a protein with an indirect, substrate-selective matrix role does **not** license a direct "protein import into the IMS" molecular function assignment. Conservation supports *plausibility of the same indirect role*, not the stronger claim encoded by the GO term.

Finding 5 — The human IBA traces to a single yeast IMP leaf

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the contradicting paper is not captured as a NOT annotation

A QuickGO provenance query pinned down the exact annotation topology. **Human HSPD1 (P10809)** has exactly **one** annotation to GO:0045041 — IBA (ECO:0000318), GO_REF:0000033, assignedBy GO_Central, qualifier `involved_in`, with/from **PANTHER:PTN000143510 + SGD:S000004249. Yeast Hsp60 (P19882 / SGD:S000004249)** has **two** annotations to GO:0045041 — the same IBA, **plus one experimental IMP** (ECO:0000315) referenced to

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 assignedBy SGD, qualifier `involved_in`.

Crucially, **no** **annotation** **cites**

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and there is **no NOT-qualified annotation** recording the contradictory stop-transfer result. The entire human annotation therefore rests on a single experimental leaf whose central claim is contested in the primary literature, and the curation record does not currently reflect that contest. This is a textbook case where phylogenetic propagation can amplify a fragile, indirect, and disputed single-organism result into a confident-looking human annotation.

Mechanistic Model / Interpretation

The core confusion the seed hypothesis rightly flags is between **chaperone contributions to translocation** and **folding after import**. Resolving that distinction is what settles the curation question.

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CYTOSOL
| precursor with bipartite b2-type presequence
v
===== OUTER MEMBRANE (TOM) =====
|
v
~~~~~ INNER MEMBRANE (TIM23) ~~~~~
|
|-- STOP-TRANSFER at inner membrane (b2 presequence)
|    --> lateral release / IMP cleavage --> IMS
|    *** Hsp60 NOT required (<a href="https://pubmed.ncbi.nlm.nih.gov/7911803/" rel="n
|
v (full matrix import, then conservative re-export)
MATRIX
|
[Hsp60/Hsp10 chaperonin]
|-- FOLDS a SUBSET of matrix proteins (rhodanese; not DHFR/barnase/Cpr3p)
|    (<a href="https://pubmed.ncbi.nlm.nih.gov/8631298/" rel="noopener noreferrer" ti
|-- ANTIFOLDING: keeps b2 unfolded to permit re-export
|    --> INDIRECT coupling of import to export (<a href="https://pubmed.ncbi.nlm.nih.g
v
IMS proteins via re-export (conservative sorting)

SEPARATE, DEDICATED IMS IMPORT PATHWAY (no Hsp60):
cysteine-rich precursors --> MIA40/CHCHD4 + Erv1 disulfide relay
--> oxidative folding traps protein in IMS (<a href="https://pubmed.ncbi.nlm.nih.gov/3392142

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Under this model, Hsp60's only documented contact with IMS biogenesis is an **indirect, matrix-side, substrate-selective antifolding role** in the conservative-sorting route (Koll 1992). That is a legitimate biological phenomenon, but it is: not translocase activity (Hsp60 does not move proteins across a membrane); not the dedicated IMS pathway (that is MIA40/CHCHD4-Erv1); contradicted on its own model substrate (b2 reaches the IMS without Hsp60 exposure); and selective (most matrix proteins fold independently of Hsp60). The GO term GO:0045041, applied with an `involved_in` qualifier to human HSPD1, reads as a direct, general involvement in IMS import — a mismatch in both specificity and confidence with what the evidence supports.

Molecular-function vs. process framing

Aspect	What the GO term implies	What the evidence actually supports
Location of action	IMS / import machinery	Mitochondrial matrix
Molecular activity	Participation in IMS translocation	ATP-dependent protein folding (chaperonin)
Substrate scope	General IMS-import substrates	Selective subset of matrix proteins
Directness	Direct involvement	Indirect antifolding coupling
Human evidence	(implied)	None — IBA only

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence & limitations
UniProt P10809 / QuickGO	Review/ database	Qualifies (weakens)	Is GO:0045041 experimentally supported in human?	Human GO:0045041 is IBA-only; all EXP terms are matrix folding; primary location = matrix	Human HSPD1	High conf on anno statu data level exper
PMID:1347713 (Koll 1992)	Mutant phenotype / direct assay	Qualifies (indirect support)	Does Hsp60 participate in IMS sorting of cyt b2?	Hsp60 "antifolds" b2 in matrix, coupling import to re- export	Yeast, isolated mitochondria	Sole leaf; role; subs spec
PMID:7911803 (Rospert/Glick 1994)	Direct assay (co-IP, translocation intermediates)	Refutes (competing)	Is Hsp60 required to sort b2- presequence proteins to IMS?	b2- presequence fusion sorted to IMS independently of hsp60; passenger never exposed to hsp60 (stop- transfer)	Yeast, isolated mitochondria	Dire cont done same subs not c as N anno
PMID:8631298 (Rospert 1996)	Direct assay	Qualifies	Do all matrix proteins need Hsp60 to fold?	Only rhodanese required hsp60; DHFR/ barnase/ Cpr3p folded without it	Yeast mitochondria	Estab subs selec

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence & limitations
PMID:9774331 (Dubaqu�� 1998)	Mutant phenotype (in vivo screen)	Qualifies	In vivo substrate spectrum of hsp60/hsp10	Overlapping but non-identical, selective substrate requirements	Yeast, in vivo	Reinforces selected
PMID:33921425 (Dickson-Murray 2021)	Review	Refutes (competing pathway)	What is the dedicated IMS import pathway?	Mia40/CHCHD4 disulfide relay is THE main IMS import pathway	Human/yeast, review	Review but could establish consistency
PMID:20136511 (Endo 2010)	Structural/review	Refutes (competing pathway)	Machinery for small soluble IMS proteins	Dedicated Tim40/Mia40-Erv1 disulfide relay	Yeast, structural	Review structural data
PMID:19477928 (Tienison 2009)	Direct assay (in vitro reconstitution)	Refutes (competing pathway)	Minimal machinery to import small Tims	Mia40 + Erv1 + O2 are sufficient; Hsp60 not involved	In vitro	Strong mechanistic evidence, alternative pathways
Sequence alignment P10809 vs P19882 (this study)	Computational (structural/evolutionary)	Qualifies (permits conservation)	Are human HSPD1 and yeast Hsp60 orthologs?	57.8% identity over 602 columns; clear orthologs	Human vs yeast	Own comparison, supports plausibility, only direct
QuickGO provenance query (this study)	Review/database	Qualifies (weakens)	What is the IBA provenance?	Single yeast IMP leaf (P 1347713); P 7911803	Human + yeast	Own database level

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence & limitations
				not recorded; no NOT		

GO Curation Implications

Lead (requires curator verification): GO:0045041 for human HSPD1 should be **down-weighted, removed as a direct function, or retained only with an explicit low-confidence / indirect-role qualifier**. It should not be presented as a core molecular activity of HSPD1.

Reasoning by GO aspect:

- **GO:0045041 is a Biological Process (BP) term.** For human HSPD1 it is supported only by IBA, and the donor evidence describes an *indirect* matrix chaperone contribution, not direct IMS translocation. The most defensible action is to treat it as **non-core**: either remove it, or keep it flagged as inference-only with a note that the underlying donor claim is contested

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 and substrate-selective
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- **The terms that should be foregrounded** are the directly supported ones: **GO:0140662 (ATP-dependent protein folding chaperone, MF)**, **GO:0042026 (protein refolding, BP, IDA)**, **GO:0050821 (protein stabilization, BP, IMP)**, and **GO:0006986 (response to unfolded protein, BP, IDA)**, with **CC = mitochondrial matrix**. These represent HSPD1's primary function.
- **If GO:0045041 is retained at all**, curators should consider linking the contradicting reference

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 — currently absent — so the IBA propagation is not silently overstated.
- Avoid defaulting to "protein binding"; the informative primary term is the chaperonin folding activity (GO:0140662), which is well supported.

Mechanistic Scope

The immediate molecular activity of HSPD1/Hsp60 that is directly evidenced is **ATP-dependent chaperonin-mediated protein folding in the mitochondrial matrix**, acting as a tetradecameric (double-heptamer) folding chamber with its co-chaperonin Hsp10 (HSPE1). Recent human structural work (e.g., [PMID:38951622](#)) confirms a conserved group I chaperonin folding cycle with client encapsulation in the folding chamber.

The IMS-import claim under evaluation is **not** this immediate activity. It is a **downstream, pathway-level consequence** observed for a specific yeast substrate (cytochrome b2) via conservative sorting, where Hsp60's contribution is to keep the substrate unfolded (antifolding) so the separate export machinery can re-translocate it. That is: direct gene-product activity = ATP-dependent protein (re)folding / holding-unfolded in the matrix; downstream/pathway consequence (not direct) = enabling IMS delivery of certain conservatively sorted substrates; not demonstrated at all for human = any IMS-import role is inferred, not measured.

Loss-of-function human phenotypes (hypomyelinating leukodystrophy, e.g., [PMID:39500555](#), [PMID:32532876](#); MitCHAP-60 disease) are attributable to impaired chaperonin assembly/folding, not to a specific IMS-import defect, and thus do not support GO:0045041 as a core function.

Conflicts and Alternatives

1. Same-substrate contradiction (strongest conflict).

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(Hsp60 couples b2 import to export) vs

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(b2-presequence fusion sorted to IMS independently of hsp60 via stop-transfer). The passenger "is never exposed to hsp60." This is a direct experimental conflict on the flagship substrate.

2. Competing dedicated pathway. The genuine, dedicated IMS import machinery is MIA40/CHCHD4-Erv1

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Hsp60 is not a component. This reframes any Hsp60 "IMS-import" role as, at most, an indirect and marginal contribution to one conservative-sorting route.

3. Substrate selectivity.

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show Hsp60's matrix role itself is not general. Generalizing a selective, indirect effect into a broad "protein import into IMS" annotation is an over-generalization.

4. **Annotation carry-over / frequency bias.** The human term is a pure IBA propagation from one yeast IMP leaf via PANTHER PTN000143510. There is no independent human confirmation, and the contradicting paper is not recorded — a classic phylogenetic over-annotation risk.
5. **Consistent alternative interpretation:** Hsp60's real biology is matrix folding; its apparent "IMS-sorting" role is a special case of its matrix antifolding/holding activity applied to a subset of conservatively sorted precursors — a downstream effect, not a distinct import function.

No paralog-confusion issue was identified (HSPD1 is the sole human mitochondrial group I chaperonin), and human/yeast orthology is solid, so the conflict is about *directness and specificity of the term*, not about identity.

Limitations and Knowledge Gaps

1. **No direct human assay either way.** What was checked: UniProt/QuickGO annotations and human structural/disease literature. Why it matters: neither a positive human IMS-import assay nor a human NOT experiment exists, so the human position is inference-bound. Resolution: a human-mitochondria import assay testing whether HSPD1 depletion affects delivery of conservatively sorted IMS substrates.

2. **Conservative sorting in humans is under-characterized.** What was checked: yeast conservative-sorting literature (cyt b2, Oxa1, COXII). Why it matters: it is unclear how prominent the conservative-sorting route is in human mitochondria versus stop-transfer and MIA pathways. Resolution: substrate-mapping of human IMS proteins by import route.
3. **The donor annotation's contest is not curated.** What was checked: QuickGO provenance. Why it matters:

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is not linked and there is no NOT annotation, so the IBA looks stronger than the evidence warrants. Resolution: curator review of the yeast leaf annotation and its counter-evidence.

4. **"Antifolding" is not cleanly captured by any single GO term.** Why it matters: the real, defensible contribution (holding a substrate unfolded to permit export) is a legitimate chaperone activity but is being represented by an import-process term. Resolution: consider whether an "unfolded protein binding"/"protein stabilization"-type framing better represents the indirect contribution than an IMS-import BP term.

Discriminating Tests

The following would most efficiently distinguish "HSPD1 directly imports IMS proteins" from "HSPD1 is a matrix folding chaperone with, at most, an indirect antifolding contribution to conservative sorting":

1. **Human HSPD1 knockdown + IMS-substrate import assay.** Deplete/inactivate HSPD1 in human cells or isolated human mitochondria and measure import/maturation of (a) MIA40/CHCHD4 substrates (small Tims, SOD1/CCS, TRIAP1) and (b) any conservatively sorted human IMS proteins. Prediction under the refutation: MIA substrates unaffected; only matrix-folding-dependent maturation affected.
2. **Co-immunoprecipitation of import intermediates** (as in

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in human mitochondria to test whether IMS-destined precursors ever physically associate with HSPD1. Prediction: bipartite/stop-transfer and MIA substrates show no stable HSPD1 binding.

3. **Route assignment of the human IMS proteome** (stop-transfer vs MIA vs conservative sorting) to quantify how many human IMS proteins could even in principle depend on a matrix Hsp60 step.
4. **Curated re-alignment of PANTHER node PTN000143510** to check whether the IBA to GO:0045041 is warranted given the contested single leaf, and whether a NOT annotation on the yeast leaf is justified.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Candidate action change: - Remove or down-weight GO:0045041 (`involved_in`) on human HSPD1, or retain only with a low-confidence, indirect-role qualifier. Rationale: IBA-only; donor evidence indirect, selective, and contested.

Candidate references to attach / verify (exact snippets): - [PMID:7911803](#): *"there was no detectable binding to hsp60 with a fusion protein that was targeted to the intermembrane space by the bipartite cytochrome b2 presequence... the cytochrome b2 presequence arrests import through the inner membrane, with the result that the attached passenger protein is never exposed to hsp60."* — counter-evidence to the donor claim; currently not linked. - [PMID:1347713](#): the sole experimental donor leaf (antifolding/conservative sorting) — verify it supports only an indirect matrix role. - [PMID:33921425](#): *"The key component of this pathway is Mia40 (called CHCHD4 in human cells)"* — establishes the true dedicated IMS-import machinery. - [PMID:8631298](#): substrate-selectivity of hsp60.

Candidate terms to foreground instead (directly supported): - GO:0140662 — ATP-dependent protein folding chaperone (MF) - GO:0042026 — protein refolding (BP, IDA) - GO:0050821 — protein stabilization (BP, IMP) - GO:0006986 — response to unfolded protein (BP, IDA) - CC: mitochondrial matrix

Suggested curator questions: - Should the yeast leaf annotation ([P 1347713](#)) carry a note/counter-reference to [P 7911803](#) ,

and should the IBA to human be re-evaluated on that basis? - Is an "antifolding/holding-unfolded" contribution better represented by a stabilization/binding term than by an IMS-import process term?

Suggested experiments: the discriminating tests above (human HSPD1-depletion IMS import assays; co-IP of import intermediates).

Bottom Line

The seed hypothesis is technically not *impossible* — human/yeast orthology is strong and an indirect antifolding contribution to conservative sorting could be conserved. But the GO term GO:0045041, as applied to human HSPD1, over-states the evidence: it is IBA-only, traces to a single contested yeast leaf, describes an indirect and substrate-selective matrix role rather than a translocase activity, and is contradicted on its own model substrate while the genuine dedicated IMS-import machinery (MIA40/CHCHD4–Erv1) is a different system. Curation should treat it as **weakly supported** / **non-core** / **inference-only** and prioritize the directly supported matrix protein-folding terms.

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